

04.03 Applying Roots and Powers

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
A power applies to a product.	Distribute the power to every factor.
A power applies to a quotient.	Apply the power to both numerator and denominator.
A power is raised to another power.	Multiply the exponents.

A. Apply powers to products

1.

$$(2x^3)^2$$

2.

$$(3a^2)^3$$

3.

$$(4y^4)^2$$

4.

$$(5m^2)^2$$

5.

$$(2p^5)^3$$

6.

$$(6b^3)^2$$

7.

$$(7q^2)^2$$

8.

$$(3r^4)^2$$

9.

$$(8s^2)^3$$

10.

$$(4t^3)^3$$

11.

$$(2uv^2)^2$$

12.

$$(5xy^3)^2$$

13.

$$(3ab^2)^3$$

14.

$$(6mn^2)^2$$

15.

$$(2c^4d)^2$$

16.

$$(4e^2f^3)^2$$

17.

$$(7gh^2)^2$$

18.

$$(3j^3k^2)^2$$

19.

$$(5l^2n)^3$$

20.

$$(2z^3w^2)^2$$

B. Apply powers to quotients

1.

$$\left(\frac{3x}{2y}\right)^2$$

2.

$$\left(\frac{2a^2}{3b}\right)^3$$

3.

$$\left(\frac{5m}{n^2}\right)^2$$

4.

$$\left(\frac{4p^3}{q}\right)^2$$

5.

$$\left(\frac{r^2}{3s}\right)^3$$

6.

$$\left(\frac{6t}{u^2}\right)^2$$

7.

$$\left(\frac{v^3}{2w}\right)^2$$

8.

$$\left(\frac{7c}{d^2}\right)^2$$

9.

$$\left(\frac{e^2}{4f}\right)^3$$

10.

$$\left(\frac{3g^4}{h}\right)^2$$

11.

$$\left(\frac{j}{5k^2}\right)^2$$

12.

$$\left(\frac{2l^3}{m}\right)^3$$

13.

$$\left(\frac{n^2}{6p}\right)^2$$

14.

$$\left(\frac{4q}{r^3}\right)^2$$

15.

$$\left(\frac{s^3}{2t^2}\right)^2$$

16.

$$\left(\frac{5u^2}{v}\right)^3$$

17.

$$\left(\frac{w}{3x^2}\right)^2$$

18.

$$\left(\frac{2y^4}{z}\right)^2$$

19.

$$\left(\frac{a^3}{4b^2}\right)^2$$

20.

$$\left(\frac{6c}{d^3}\right)^2$$

C. Raise powers to powers

1.

$$(x^4)^3$$

2.

$$(a^2)^5$$

3.

$$(y^3)^4$$

4.

$$(m^5)^2$$

5.

$$(p^2)^6$$

6.

$$(b^4)^2$$

7.

$$(q^3)^3$$

8.

$$(r^2)^7$$

9.

$$(s^5)^3$$

10.

$$(t^3)^5$$

11.

$$(u^4)^4$$

12.

$$(v^2)^8$$

13.

$$(w^3)^6$$

14.

$$(c^5)^2$$

15.

$$(d^2)^9$$

16.

$$(e^4)^3$$

17.

$$(f^3)^7$$

18.

$$(g^2)^6$$

19.

$$(h^5)^4$$

20.

$$(j^3)^8$$